ASCII numbers

ASCII stands "American Standard Code for Information Interchange". Although ASCII is standard in the United States (and many other countries), BCD numbers have universal application. However, the keyboards, printers, and monitors all use ASCII. It is, therefore, helpful to know the conversion procedure of ASCII to BCD and vice versa. Table 8.2 shows the list of Keyboards keys (in ASCII), the corresponding binary number and unpacked BCD codes.

Table 1 Keyboard Keys with ASCII, Binary, and BCD representation

Key	ASCII (hex)	Binary	BCD (unpacked)
0	30	011 0000	0000 0000
1	31	011 0001	0000 0001
2	32	011 0010	0000 0010
3	33	011 0011	0000 0011
4	34	011 0100	0000 0100
5	35	011 0101	0000 0101
6	36	011 0110	0000 0110
7	37	011 0111	0000 0111
8	38	011 1000	0000 1000
9	39	011 1001	0000 1001

ASCII to BCD Conversion:

It is necessary to convert the ASCII data provided by the keyboard into BCD, before processing the data in BCD. There are instructions which require unpacked BCD while for other instructions packed BCD representation is required. Here, each conversion is discussed separately.

ASCII to Packed BCD Conversion:

The conversion process for ASCII to packed BCD is shown below step by step.
Consider the number "49" is input using the keyboard and is to be converted in to packed BCD

The number stored as **3439** in the memory.
Key "4" is pressed first, the ASCII code for "4" is "34h" i.e., $0011\ 0100_b$
Key "9" is pressed then, the ASCII code for "9" is "39h" i.e., $0011\ 1001_b$

1. Remove "3" from both the numbers by masking the upper nibble as follows

```
       0011 0100
(AND)  0000 1111
       0000 0100  →  "04"······ (key1)
```

Similarly,

```
       0011 1001
(AND)  0000 1111
       0000 1001  →  "09"······ (key2)
```

Note here that the key1 and key2 show the numbers is **Unpacked BCD form**.

2. Shift Left the 4 bits of key1, it will result in 0100 0000

3. Perform OR operation to 0100 0000 with key2

```
        0010 0000
(OR)    0000 1001
        0100 1001  →  "49"
```

Packed BCD to ASCII Conversion:
The conversion process for packed BCD to ASCII is shown below step by step.
Consider the BCD number "49" is to be converted in to ASCII

1. Copy the BCD number "49" in two 8 bit variables, var1 and var2

2. var1 = 0100 1001, var2 = 0100 1001

3. Mask lower nibble from var1 and upper nibble from var2

```
        var1                  var2
        0010 1001             0010 1001
(AND)   1111 0000     (AND)   0000 1111
        0010 0000             0000 1001
```

4. Shift Right the 4 bits of var1 to make it "0000 0100"
   Now, var1 = "0000 0100" (Unpacked BCD form)
   And var2 = "0000 1001" (Unpacked BCD form)

5. Perform OR operation to var1 with 30h and OR operation to var2 with 30h

```
        var1                          var2
        0000 0100                     0000 1001
(OR)    0011 1111             (OR)    0011 0000
        0011 0100  →  "34"            0011 1001  →  "39"
```

NOTE:

ASCII to BCD conversion => ASCII → Unpacked BCD → Packed BCD
BCD to ASCII conversion => Packed BCD → Unpacked BCD → ASCII